

VOICE INTEGRATION · ARCHITECTURE COMPARISON

Handy × agent-meow × Hermes vs Cherry AI

A deep-dive comparison of our open-source voice agent stack against Cherry's cloud-credit model — covering STT, TTS, system control, and the full voice I/O loop.

4

COMPONENTS

11

TESTS PASSING

2

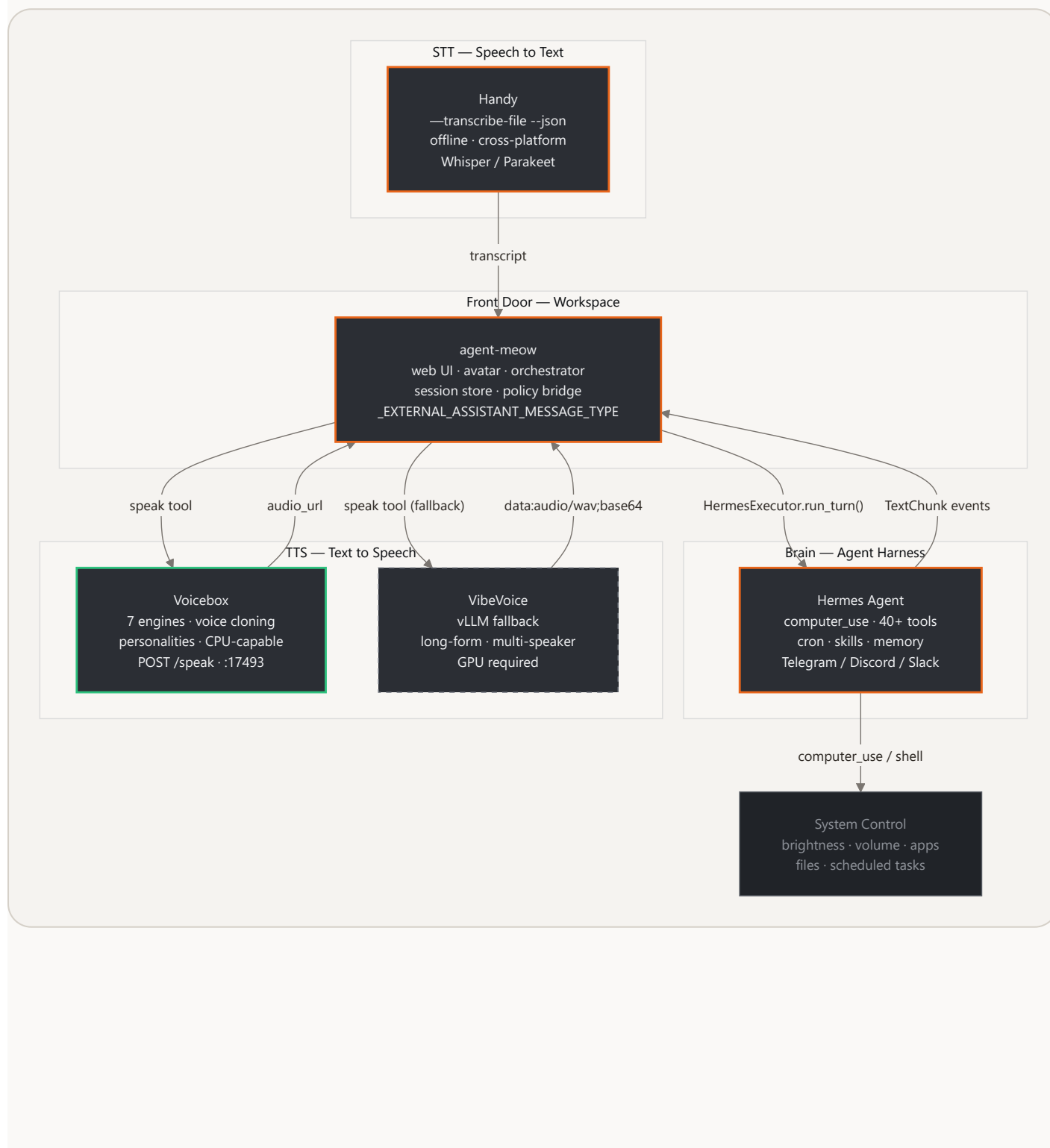
FILES CHANGED

0

RECURRING COST

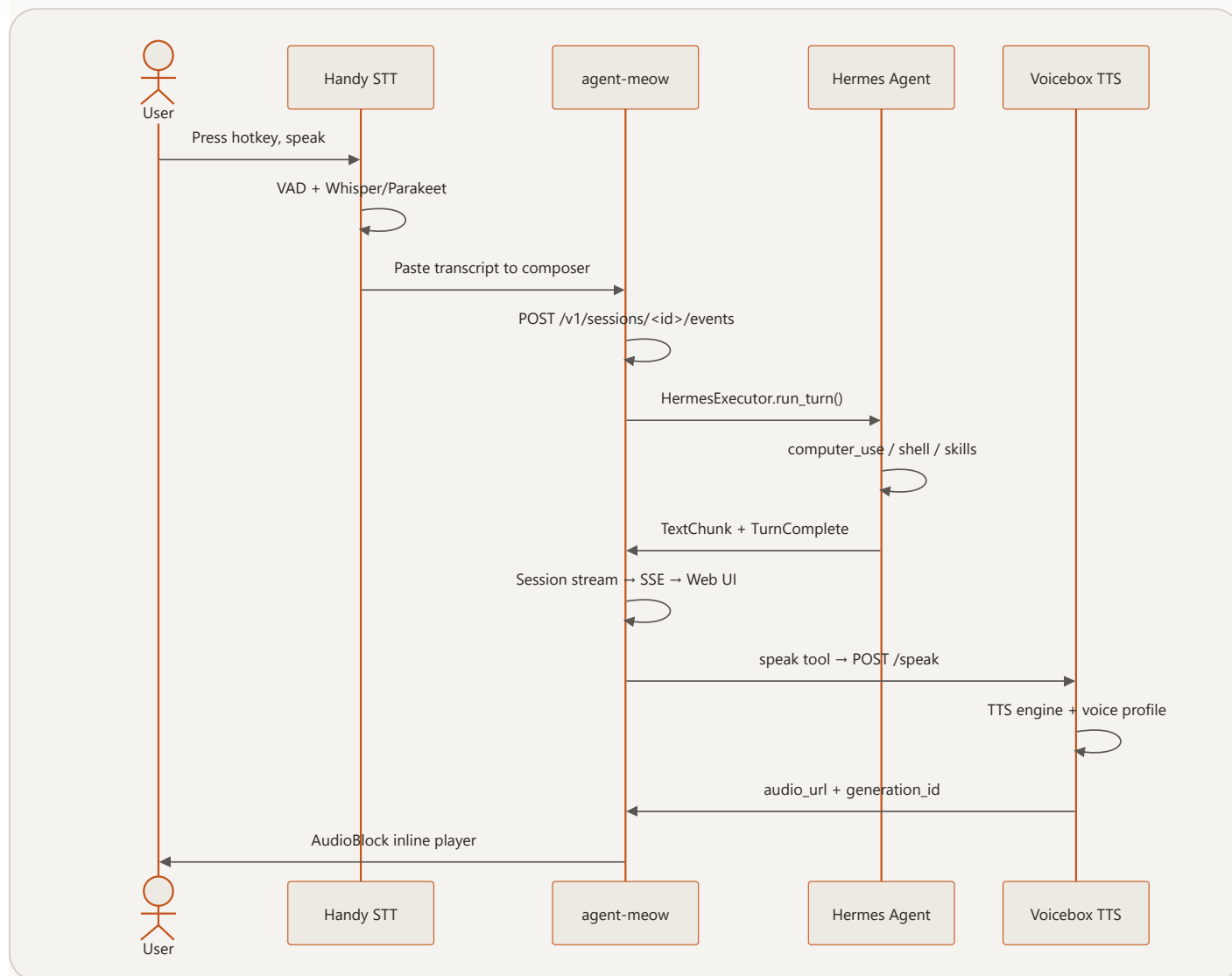
Our Stack Architecture

Four open-source components, each doing one thing well. Fully offline, cross-platform, no cloud credits.



The Full Voice I/O Loop

Speak → Handy transcribes → agent-meow receives → Hermes executes → Voicebox speaks the response. All local, all offline.



Capability Comparison Matrix

Feature-by-feature comparison across every dimension that matters for a voice-controlled agent platform.

CAPABILITY	OUR STACK	CHERRY	VERDICT
STT (voice input)	Handy — offline, cross-platform, Whisper/Parakeet, --transcribe-file --json	Cloud ASR (28ms claim)	We win
TTS (voice output)	Voicebox (7 engines, cloning, personalities) + VibeVoice (vLLM fallback)	Cloud TTS	We win
Agent loop	Hermes — 40+ tools, computer_use, cron, skills, memory, subagents	30+ tools, multi-AI coordination	Tie
System control	computer_use (set_value on sliders) + shell (WMI/PowerShell)	系统控制 — brightness, volume, apps	Tie
Desktop automation	computer_use — background, no cursor steal, cross-platform	模拟鼠标点击控制	We win
Scheduled tasks	cronjob_tools — cron + platform delivery	定时任务	Tie
Messaging	Telegram / Discord / Slack / WhatsApp / Signal (Hermes gateway)	微信互联 (WeChat)	Cherry wins
Workspace / UI	agent-meow web UI — avatar, panels, docs, images, videos, terminals, subagents	语音控制工作台	We win
App management	shell / computer_use (winget/choco launch)	应用商店 — dedicated app store	Cherry wins
File management	file_operations + agent-meow Files panel	我的文件	Tie
PC diagnostics	Not built (deprioritized)	电脑医生 — system health monitor	N/A
Local LLM	Hermes (any endpoint incl. local Ollama) + Handy custom provider	Cloud credits only	We win

CAPABILITY	OUR STACK	CHERRY	VERDICT
Cross-platform	Windows / macOS / Linux (Handy + Hermes)	Windows AIPC only	We win
Privacy	Fully offline — Handy STT + local LLM + Voicebox TTS	Cloud-dependent	We win
OEM branding	agent-meow branded front door + Voicebox voice cloning	Cherry-branded	Tie
Pricing	Free — open source, local compute	¥0–129/month — credits expire in 30 days	We win

Score Summary

Head-to-head tally across all 16 capabilities.

W

We win

Offline STT, TTS flexibility, desktop automation, workspace UI, local LLM, cross-platform, privacy, pricing

T

Tie

Agent loop, system control, scheduled tasks, file management, OEM branding

C

Cherry wins

WeChat integration, app store polish

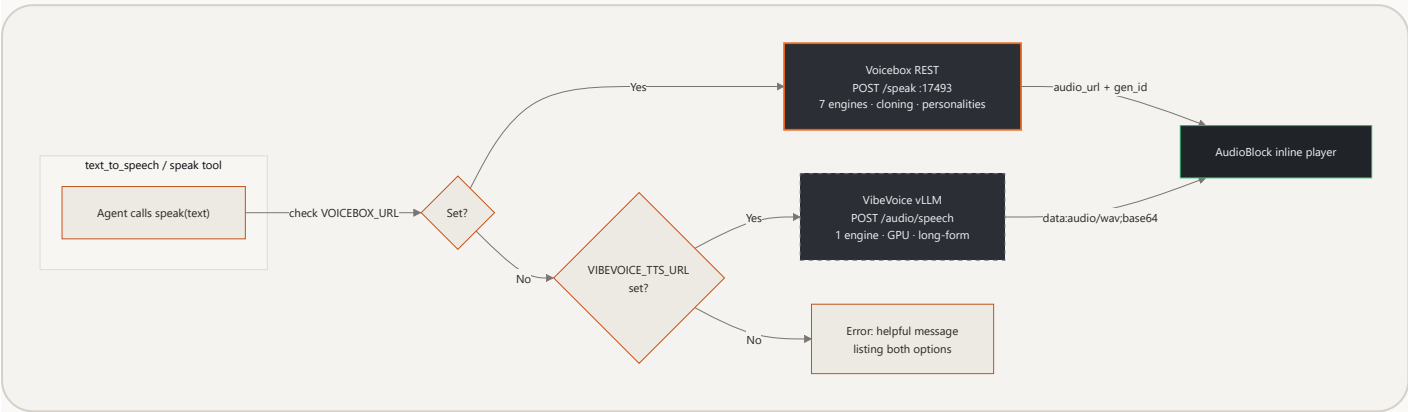
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Not applicable

PC diagnostics — deprioritized by design decision

TTS Integration — Two Paths

The `_execute_voice_tool` function now supports dual TTS routing: Voicebox (preferred) and VibeVoice (fallback).



FEATURE	VOICEBOX (PREFERRED)	VIBEVOICE (FALLBACK)
Env var	VOICEBOX_URL	VIBEVOICE_TTS_URL
TTS engines	7 (Qwen3-TTS, LuxTTS, Chatterbox, TADA, Kokoro...)	1 (diffusion-based, Qwen2.5 backbone)
Voice cloning	Yes — 5 cloning engines, 3–30s sample	No
Personalities	Yes — per-profile persona rewrite via local LLM	No
GPU required	No — Kokoro (82M, CPU) + LuxTTS (150x CPU)	Yes — vLLM + GPU
Long-form	Auto-chunking with crossfade	Up to 90 min, 4 speakers
Returns	audio_url (pollable) + generation_id + profile	data:audio/wav;base64 (inline)

Code Change Summary

Two files changed in agent-meow. The rest of the git diff is omnigent→agent-meow rebrand noise.

```
# File 1: agent_meow/runner/tool_dispatch.py # Added VOICEBOX_URL support to
_execute_voice_tool() if voicebox_url: # Route through Voicebox REST API (POST /speak)
payload = {"text": text} if profile: payload["profile"] = profile if engine:
payload["engine"] = engine resp = await server_client.post( f"{voicebox_url}/speak",
json=payload, timeout=120.0 ) # Returns: audio_url + generation_id + profile + source:
"voicebox" elif tts_url: # Fallback: VibeVoice-TTS via vLLM gateway resp = await
server_client.post( f"{tts_url}/audio/speech", json=payload, timeout=120.0 ) # Returns:
data:audio/wav;base64 + source: "vibevoice" # File 2:
tests/runner/test_voice_tool_dispatch.py (new, 11 tests) # All passing: 11 passed in
0.24s
```

Implementation Status

What's code-ready vs what needs wiring. Most of the stack was already built — we only added the Voicebox TTS path.

- **Handy --transcribe-file** — fully implemented in Handy main. Outputs JSON with text, model, rtf. Headless mode bypasses GUI.
- **_execute_voice_tool** — fully implemented in tool_dispatch.py. Handles Handy STT + Voicebox/VibeVoice TTS. 11 tests passing.
- **_VOICE_TOOLS frozenset** — all 4 tool names in _ALL_LOCAL_TOOLS + _NATIVE_RELAY_BUILTIN_TOOLS. Dispatch routing at line ~5901.
- **HermesExecutor** — agent-meow drives Hermes as a harness with session resume (--resume session_id) + policy bridge (pre_tool_call hook).
- **_EXTERNAL_ASSISTANT_MESSAGE_TYPE** — sessions.py has built-in mechanism for landing external agent results into the workspace transcript.
- **AudioBlock.tsx + parseAudioFromToolOutput()** — component exists, parses audio_url from JSON tool output. NOT yet wired into ToolCard.
- **AudioBlock → ToolCard wiring** — Task 4: ~10 lines in ToolCard.tsx to call parseAudioFromToolOutput before rendering raw text.
- **Voicebox MCP server config** — Task 1: ~10 lines YAML for stdio transport MCP server registration.
- **voicebox-agent example spec** — Task 2: ~30 lines YAML example agent with Voicebox MCP + Handy STT tools.

Remaining Gaps vs Cherry

Explicitly deprioritized per design decisions. Listed here for completeness.

W

WeChat integration

Hermes has 5 messaging platforms (Telegram, Discord, Slack, WhatsApp, Signal). WeChat would require a new gateway adapter — China-market specific.

Not built**A**

App store surface

Cherry has a dedicated app store UI. Our stack uses shell/computer_use (winget/choco) — functional but less polished as a consumer surface.

Not built**D**

PC diagnostics

Cherry's 电脑医生 monitors system errors, network, drivers, device health. We deprioritized this — Hermes could add it as a skill (cron + WMI/EventLog).

Deprioritized

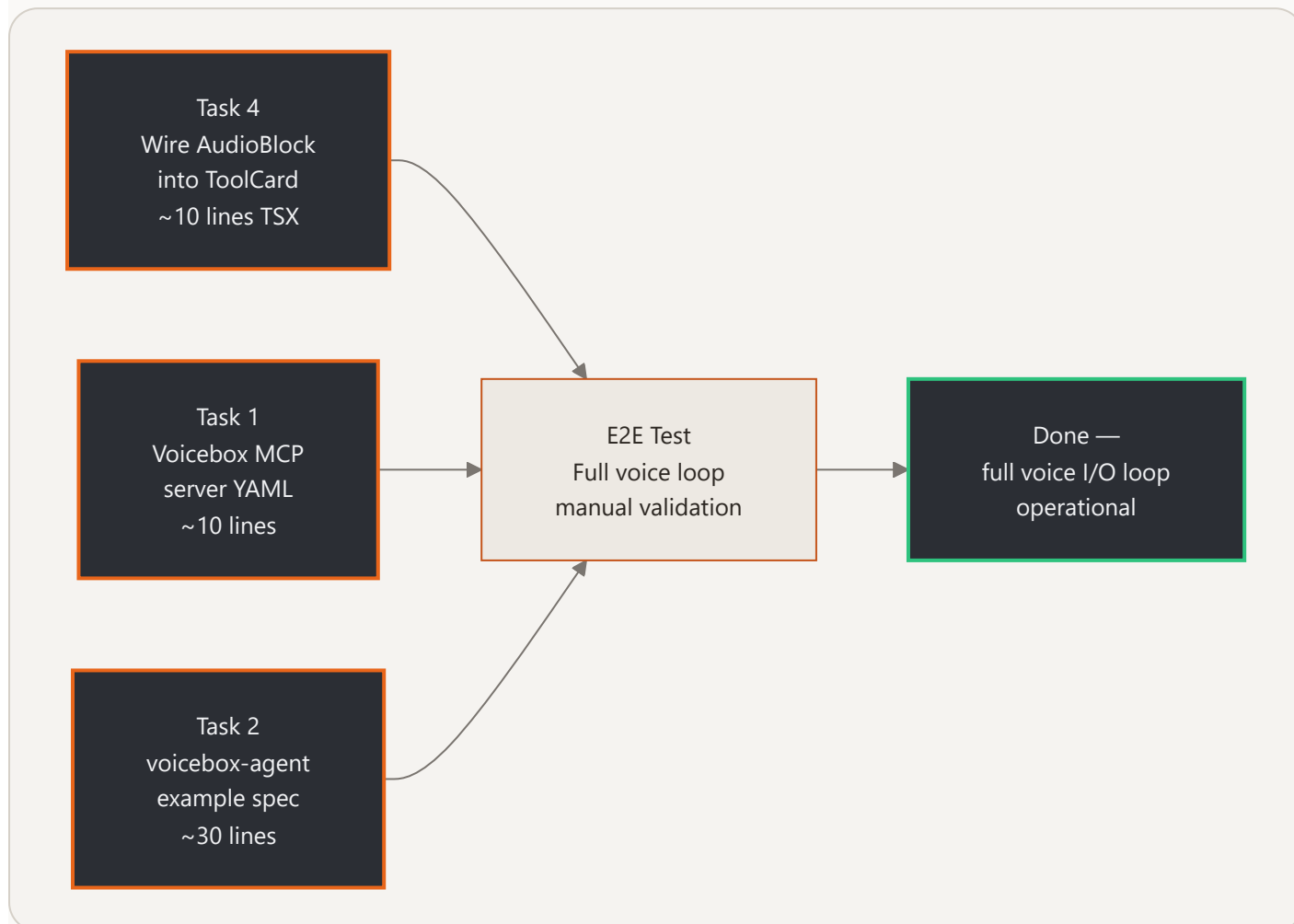
Pricing Model Comparison

Our stack is free and local. Cherry charges monthly credits that expire in 30 days.

TIER	CHERRY PRICE	CHERRY CREDITS	OUR STACK
Free	¥0	30,000	Everything — no limits
Basic	¥69/month	600,000 (expires 30 days)	—
Advanced	¥99/month	900,000 (expires 30 days)	—
Super	¥129/month	1,200,000 (expires 30 days)	—
Annual cost	¥0–1,548	Credits expire monthly	¥0 — forever

Next Steps

Three small tasks remaining to complete the voice integration.



4

Wire AudioBlock into ToolCard

Import `parseAudioFromToolOutput` in `ToolCard.tsx`, check for `audio_url` before rendering raw JSON. ~10 lines.

~10 lines TSX

1

Voicebox MCP server config

Create agent_meow/tools/mcp/voicebox.yaml with stdio transport. ~10 lines YAML.

~10 lines YAML

2

voicebox-agent example spec

Create examples/voicebox-agent/config.yaml with Voicebox MCP + Handy STT. ~30 lines YAML.

~30 lines YAML

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Voicebox